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import numpy as np

# Example for IIT / Phi computation with a tiny binary neural network.
# Requires: pip install pyphi
import pyphi

# Transition probability matrix (TPM)
# State order for 3 binary nodes: 000,001,010,011,100,101,110,111
# Here each row gives the next-state probabilities for each node.
tpm = np.array([
    [0, 0, 0],
    [0, 0, 1],
    [0, 1, 0],
    [0, 1, 1],
    [1, 0, 0],
    [1, 0, 1],
    [1, 1, 0],
    [1, 1, 1],
])

# Connectivity matrix
cm = np.array([
    [0, 1, 1],
    [1, 0, 1],
    [1, 1, 0],
])

labels = ('A', 'B', 'C')
network = pyphi.Network(tpm, cm=cm, node_labels=labels)

# Current system state
state = (1, 0, 1)
subsystem = pyphi.Subsystem(network, state)

# Compute small phi for the subsystem
phi_value = pyphi.compute.phi(subsystem)
print('Phi:', phi_value)

# Compute the major complex and its big Phi
sia = pyphi.compute.major_complex(network, state)
print('Major complex:', sia.subsystem)
print('Big Phi:', sia.phi)

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